
Hospital Resource Management Database System

Relational Database Design for Patients, Doctors, Rooms, Appointments, and Admissions

Project Title	Hospital Resource Management Database System
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Abstract

This paper presents the design and implementation of a relational database system for managing core hospital resources: patients, doctors, rooms, appointments, and admissions. Manual or unstructured record-keeping in healthcare settings is prone to lost records, duplicated data, slow retrieval, and poor overall management — problems a relational Database Management System (DBMS) is specifically designed to solve. This paper documents the five core entities of the system, their relationships, the SQL schema used to implement them with primary and foreign key constraints, representative queries including a multi-table JOIN, and the normalization principles applied to keep the schema free of redundancy and update anomalies.

Keywords: DBMS, relational database, SQL, primary key, foreign key, normalization, JOIN, healthcare data management

1. Introduction

Hospitals generate and depend on large volumes of structured information every day: patient demographics, doctor schedules, room occupancy, appointment bookings, and admission records. Without a database system, this information is typically scattered across paper records or disconnected spreadsheets, which leads directly to lost records, duplicated entries, slow retrieval during time-critical situations, and generally poor operational management. A relational DBMS addresses these problems by organizing information into structured, related tables governed by well-defined integrity constraints. This project, undertaken as a Semester 4 mini project under Database Management Systems, designs and implements such a system for a simplified hospital resource management scenario.

2. Database Entities

The system is built around five core entities. The Patient entity stores patient demographic and contact details. The Doctor entity stores staff information including medical specialization and years of experience. The Room entity tracks physical room resources and their current occupancy status. The Appointment entity records scheduled meetings between a specific patient and doctor. The Admission entity records hospitalization events, linking a patient to an assigned room. Together, these five entities and their relationships model the complete patient journey from registration through appointment booking to potential hospital admission.

3. Entity Relationship Structure

Figure 1 illustrates the relationship flow between entities. A patient is linked to one or more appointments; each appointment is linked to the doctor conducting it. Separately, a patient may be linked to an admission record, which in turn is linked to the room assigned to that admission. This structure keeps each entity's data isolated in its own table while foreign keys preserve the relationships between them, avoiding the redundancy that would result from storing doctor or room details directly inside the patient record.

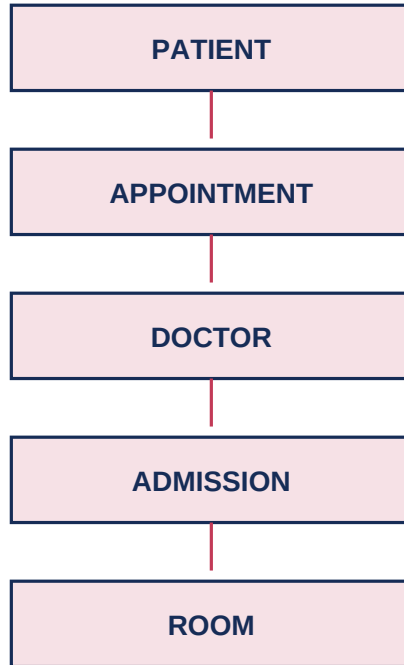


Figure 1. Entity relationship flow across the five core tables of the hospital database.

4. Database Schema

Table 1 summarizes the field structure of each of the five tables. Every table defines a single-column primary key that uniquely identifies its rows; the Appointment and Admission tables additionally define foreign keys that reference the Patient, Doctor, and Room tables to enforce referential integrity between related records.

Patient	PatientID (PK), Name, Age, Gender, Phone
Doctor	DoctorID (PK), Name, Specialization, Experience
Room	RoomID (PK), RoomType, Status
Appointment	AppointmentID (PK), PatientID (FK), DoctorID (FK), Date
Admission	AdmissionID (PK), PatientID (FK), RoomID (FK), AdmissionDate

Table 1. Field structure of the five core tables. PK = Primary Key, FK = Foreign Key.

5. SQL Implementation

The schema is implemented directly in standard SQL. The Patient, Doctor, and Room tables are independent and define only a primary key each. The Appointment and Admission tables additionally declare foreign key constraints that tie each record back to the patient, doctor, and room it concerns, ensuring the database rejects any appointment or admission that references a patient, doctor, or room that does not exist.

```

CREATE TABLE Appointment (
  AppointmentID INT PRIMARY KEY,
  PatientID INT,
  DoctorID INT,
  FOREIGN KEY (PatientID) REFERENCES Patient(PatientID),
  FOREIGN KEY (DoctorID) REFERENCES Doctor(DoctorID)
);

CREATE TABLE Admission (
  AdmissionID INT PRIMARY KEY,
  PatientID INT,
  RoomID INT,
  FOREIGN KEY (PatientID) REFERENCES Patient(PatientID),
  FOREIGN KEY (RoomID) REFERENCES Room(RoomID)
);

```

Listing 1. Foreign key constraints linking Appointment and Admission to their referenced tables.

6. Representative Queries

Beyond simple selection queries such as listing all patients or filtering rooms by availability status, the system supports multi-table JOIN queries that reconstruct a complete view of a scheduled visit by combining data from the Appointment, Patient, and Doctor tables in a single result set, as shown in Listing 2.

```

SELECT Patient.Name, Doctor.Name
FROM Appointment
JOIN Patient ON Appointment.PatientID = Patient.PatientID
JOIN Doctor ON Appointment.DoctorID = Doctor.DoctorID;

```

Listing 2. JOIN query combining patient and doctor details for each scheduled appointment.

7. Database Workflow

Operationally, the system models a five-stage patient journey: patient registration creates a row in the Patient table; doctor assignment and appointment booking create a corresponding row in the Appointment table referencing both the patient and the assigned doctor; if hospitalization is required, an admission record is created; and room allocation updates the Room table's status field to reflect occupancy. This workflow ensures every stage of patient care is captured as a discrete, queryable database event.

8. Normalization

The schema was designed to satisfy the first three normal forms. First Normal Form (1NF) is satisfied because every field holds a single atomic value with no repeating groups — for example, a patient with multiple appointments has multiple rows in the Appointment table rather than a single row with a repeated appointment field. Second Normal Form (2NF) is satisfied because every non-key attribute depends on the whole of its table's primary key; since each table here uses a single-column primary key, no partial dependency is possible. Third Normal Form (3NF) is satisfied because no non-key attribute depends transitively on another non-key attribute — for instance, a doctor's specialization is stored once in the Doctor table and referenced by ID from Appointment, rather than being duplicated into every appointment row.

9. Applications

A relational hospital database of this design applies directly to hospitals for patient management, clinics for doctor scheduling, medical centers for room allocation, emergency services for admission tracking, and the broader healthcare industry for maintaining structured medical records across departments.

10. Result

A Hospital Resource Management Database was successfully designed and implemented using DBMS concepts. The system manages patient records, doctor information, room allocation, appointments, and admissions through five structured relational tables connected by primary and foreign keys, and supports both single-table filtering and multi-table JOIN queries. The accompanying interactive HTML tool reproduces the schema and query behavior described in this paper against representative sample data.

References

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